

Multiple Choice (10 marks)

1. The difference between dc and ac in electric circuits is that in dc, charges flow
 - (a) to and fro
 - (b) in the direction opposite to the flow of current
 - (c) in a circular direction
 - (d) only when the circuit is closed
 - (e) in a random direction
2. A camera objective of 50 mm focal length is set at aperture stop ($f / 4$). If the camera is then focused at an object 20 cm away what will the effective f-number be?
 - (a) 7.1
 - (b) 9.4
 - (c) 4.0
 - (d) 3.8
 - (e) 5.33
3. What is the second most common element in the solar system?
 - (a) Methane
 - (b) Nitrogen
 - (c) Oxygen
 - (d) Iron
 - (e) Hydrogen
4. Colors in a soap bubble result from light
 - (a) dispersion
 - (b) deflection
 - (c) refraction
 - (d) reflection
 - (e) interference
5. An electric iron with a resistance of 24 ohms is connected to a 120-volt line. What power is being used?
 - (a) 600 watts
 - (b) 800 watts

- (c) 750 watts
- (d) 1000 watts
- (e) 700 watts

True / False (10 marks)

Answer True or False

6. True or False: In an ideal pulley system, pulling 1 m of rope down will always raise the crate by the same distance of 1 m.
7. True or False: A magnetic force acting on a beam of electrons can change the direction of their motion without altering their charge.
8. True or False: The minimum thickness of crystalline quartz required for a quarter wave plate at 589 nm is 0.016 mm.
9. True or False: A freight engine of 400 hp can pull a string of freight cars weighing 200 tons at a speed of 75 mi/hr, considering rolling friction.
10. True or False: When the temperature of a strip of iron is increased, its length decreases due to thermal contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A₁₀₀-kg man jumps from a height of 5 m into a swimming pool, with the water taking 0.4 seconds to bring him to rest. Calculate the average force exerted by the water and discuss the underlying physics principles involved.
12. Calculate the time it takes for a meter stick traveling at 0.8 c to pass an observer. Discuss the implications of relativistic effects on measurements in different reference frames.

End of Paper